arXiv:2310.01208v1 [cs.CL] 2 Oct 2023

LABEL SUPERVISED LLAMA FINETUNING*

A PREPRINT

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ABSTRACT

The recent success of Large Language Models (LLMs) has gained significant attention in both academia and industry. Substantial efforts have been made to enhance the zero- and few-shot generalization capabilities of open-source LLMs through finetuning. Currently, the prevailing approach is instruction-tuning, which trains LLMs to complete real-world tasks by generating responses guided by natural language instructions. It is worth noticing that such an approach may underperform in sequence and token classification tasks. Unlike text generation tasks, classification tasks have a limited label space, where precise label prediction is more appreciated than generating diverse and human-like responses. Prior research has unveiled that instruction-tuned LLMs cannot outperform BERT, prompting us to explore the potential of leveraging latent representations from LLMs for supervised label prediction. In this paper, we introduce a label-supervised adaptation for LLMs, which aims to finetuning the model with discriminant labels. We evaluate this approach with Label Supervised LLaMA (LS-LLaMA), based on LLaMA-2-7B, a relatively small-scale LLM, and can be finetuned on a single GeForce RTX4090 GPU. We extract latent representations from the final LLaMA layer and project them into the label space to compute the cross-entropy loss. The model is finetuned by Low-Rank Adaptation (LoRA) to minimize this loss. Remarkably, without intricate prompt engineering or external knowledge, LS-LLaMA substantially outperforms LLMs ten times its size in scale and demonstrates consistent improvements compared to robust baselines like BERT-Large and RoBERTa-Large in text classification. Moreover, by removing the causal mask from decoders, LS-unLLaMA achieves the state-of-the-art performance in named entity recognition (NER). Our work will shed light on a novel approach to adapting LLMs for various downstream tasks.

1 Introduction

Large Language Models (LLMs), such as GPT-3 [Brown et al., 2020] and GPT-4 from OpenAI, LLaMA [Touvron et al., 2023a,b] from Meta, and PaLM [Chowdhery et al., 2022, Anil et al., 2023] from Google, have demonstrated impressive language understanding and human-like response generation abilities. The large-scale pretraining and increased parameter size lead to phenomenal *emergent abilities* [Wei et al., 2022a], endowing LLMs with strong generalization capacity for unseen tasks, even in zero- and few-shot settings. Such capabilities of LLMs, unseen in smaller models, have significantly revolutionized methodologies across various natural language processing (NLP) tasks, ranging from label-supervised parameter optimization to prompt-based response generation. A key driving factor behind these transformations is the decoder-only architecture of these LLMs, which incorporates causal masks to prevent forward information exposure. Consequently, until now, the latent representations of LLMs have primarily been employed for predicting the next token in autoregressive text generation.

*Preprint. Work in progress.

LLMs have been extensively evaluated in text classification and information extraction tasks, both in zero- and few-shot settings [Zhao et al., 2021, Wei et al., 2023, Li et al., 2023a], as well as with instruction tuning Wang et al. [2023], Lei et al. [2023]. Recent studies have revealed that zero-shot LLMs struggle to achieve satisfactory performance in these domains. For instance, GPT-3 (175B) achieves a classification accuracy of only 76% on SST-2 and 43.9% on AGNews [Zhao et al., 2021], while GPT-3.5-Turbo (154B) attains an F1 score of 18.22% for named entity recognition (NER) on OntoNotes [Wang et al., 2023]. Even the popular ChatGPT can only achieve F1 scores of 67.2% and 51.1% for NER on CoNNL2003 and OntoNotes, respectively Li et al. [2023a]. The above results fall short of state-of-the-art benchmarks. Although metric performance can be enhanced through careful prompt engineering and instruction-tuning techniques, these refined LLMs can hardly outperform discriminative encoder models such as BERT [Devlin et al., 2019] and RoBERTa [Liu et al., 2019].

We consider that these observations align with expectations, as LLMs and BERT possess different architectures and strengths that lead to divergent performance on specific tasks like text classification and NER. In classification tasks, labels often consist of highly concentrated words or phrases, and NER tasks involve symbolic tags, causing difficulties for LLMs in understanding the semantic meanings of the labels effectively. Although conversational LLMs can be finetuned on labeled datasets for these tasks, their generation-focused architecture may not capture task-specific patterns as efficiently as label-supervised BERT models, which have consistently demonstrated superior performance on a wide range of label prediction tasks. Furthermore, label spaces are considerably more restricted compared to the entire vocabulary, making the autoregressive generation less effective and efficient in label prediction, particularly in NER, where outputs are structured sequences of NER tokens.

Motivated by the efficacy of finetuning BERT and RoBERTa for classification tasks, we are the first to explore the feasibility of finetuning LLMs with label supervision to achieve effective task-specific adaptation. In this study, we employ the open-sourced LLaMA-2-7B model. We directly extract latent vector representations from the final LLaMA decoder layer, which was originally designed for autoregressive next-token prediction. These representations are then mapped into the label space through feed-forward layers, yielding logits that are used for discriminant label classification. We calculate the cross-entropy loss and employ Low-Rank Adaptation (LoRA) [Hu et al., 2021] to fine-tune the LLaMA model. Our preliminary results on multiclass benchmarks have been remarkably promising. We have observed significant improvements over both zero-shot and instruction-tuned LLaMA-2-7B models, as well as consistent enhancements compared to finetuned BERT. Extensive experiments demonstrate the effectiveness and robustness of our proposed task-specific LLaMA adaptation. We name such a configuration **L**abel **S**upervised **LLaMA** (LS-LLaMA).

Before this work, open-sourced LLMs had yet been utilized for discriminant label classification as their latent representations were considered not suitable for language encoding, although Meta has provided the interface for sequence classification since LLaMA-1[2] [Touvron et al., 2023a]. Unlike the the encoder-based structure of Transformer [Vaswani et al., 2017] models, such as BERT and RoBERTa, conversational LLMs do not have an encoder structure Brown et al. [2020], Touvron et al. [2023a] and were trained primarily for language generation, focusing on predicting the next word in a sequence. The causal mask in the decoder blocks avoids forward information disclosure and also blocks bidirectional dependency extraction, leading to a myth that LLMs are not suitable for text encoding. Our research has has unveiled that the decoder output of LLMs contains a substantial amount of semantic meaning from the input text and can be effectively utilized as text representations for various classification tasks. However, the causal mask's presence causes fatal information loss at the token-level representation, leading to unsatisfactory results by LS-LLaMA in NER tasks. To overcome such a limitation, we propose an innovative solution by removing the causal mask from the decoders. After finetuning, the **L**abel **S**upervised **un**masked **LLaMA** (LS-unLLaMA) exhibits substantial improvements (up to 18%) over LS-LLaMA in NER benchmarks i.e., CoNNL2003 and OntoNotes, and even outperforms LS-LLaMA on multiple text classification tasks.

This research aims to introduce a label-supervised adaptation configuration for LLMs. We investigate the feasibility of employing latent representations from LLMs for discriminant label prediction in text classification tasks. Our experimental results demonstrate that LLMs' latent representations can effectively serve as a text encoding method. Furthermore, we identify a limitation when using LLM's representations for token-level applications, attributed to the causal masks in the decoder structure, which restrict bidirectional information flow. To overcome this limitation, we remove the causal masks from LLaMA, resulting in state-of-the-art performance in NER tasks.

[2] `https://huggingface.co/docs/transformers/v4.33.2/model_doc/llama#transformers.LlamaForSequenceClassification`

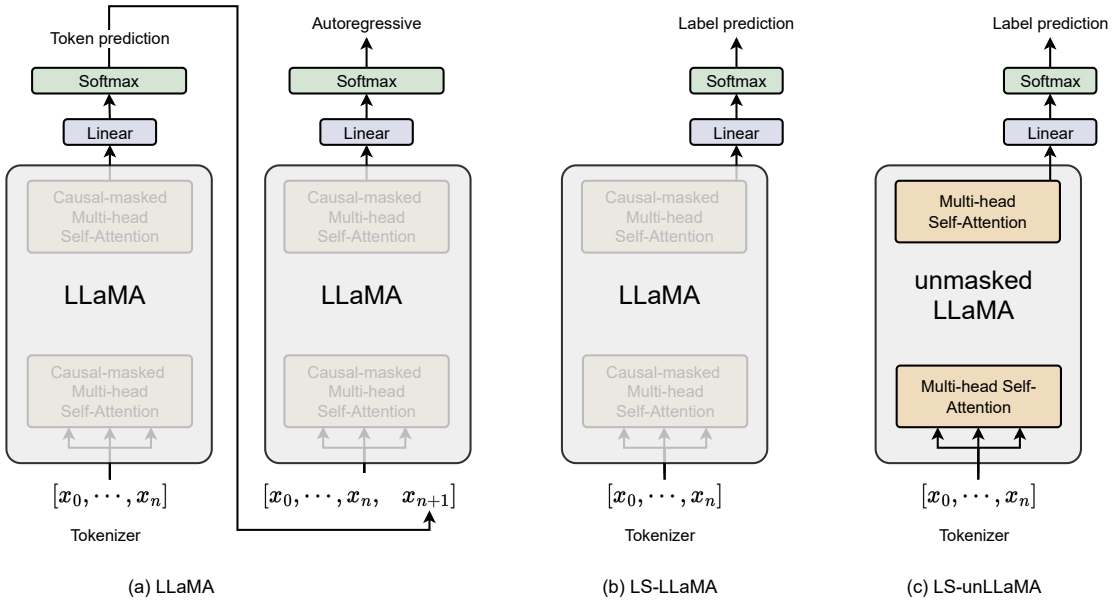

Figure 1: Comparison between conversational LLaMA and our proposed LS-LLaMA and LS-unLLaMA in sequence classification.

2 Methodology

This section presents the implementation of our proposed LS-LLaMA and LS-unLLaMA. Figure 1 depicts how they are different from the conventional autoregressive settings.

2.1 Label-supervised LLaMA

We tokenize the input sequence S with the default `AutoTokenizer` following the automatic operation of the transformers[3] library. The tokens T were fed into pretrained models through `LlamaForSequenceClassification` to extract the latent representation H from LLaMA for sequence classification,

$$\begin{aligned} T &= \text{Tokenizer}(S) \\ H_{seq} &= \text{LlamaForSeqClf}(T). \end{aligned} \tag{1}$$

The pooling operation is applied to the latent representation to obtain the vector representation h for sequence classification. `LlamaForSequenceClassification`'s default pooling operation takes out the last token vector from the final representation, as this is the only token that encodes all the historical information due to single-direction information flow caused by the causal masks. After passing through fully connected layers and a softmax layer, vector representation h is mapped to the label space. Cross-entropy loss is calculated based on the output logits and the ground-truth label.

LLaMA did not provide the interface for token classification. Therefore, we modify the LLaMA model to obtain all the token representations with `LlamaForTokenClassification`[4], $H_{tkn} = \text{LlamaForTokenClf}(T)$, for token classification tasks.

We apply LoRA [Hu et al., 2021] to finetune the LLaMA model to maximize the probability of the correct label.

[3] `https://github.com/huggingface/transformers`

[4] Code for LS-LLaMA is available on GitHub: `https://github.com/4AI/LS-LLaMA`